

All-order localized derivative bounds for the Laguerre Poisson kernel

Auxiliary analytic argument; independent mathematical reviews completed

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The full argument passed separate mathematical reviews by the size-bounds and rational-design collaborators. This is not formal verification. This note establishes an analytic input for a possible improvement of endpoint-profile bounds. It does not establish a new endpoint-profile rate: polynomial truncation, integration over the row measure, and the final monotonicity argument remain separate tasks.

For $0 < \tau < 1$ define the positive Laguerre Poisson kernel

$$K_\tau(x, a) = \frac{1}{\tau} \exp \left[-\frac{(1-\tau)(x+a)}{\tau} \right] I_0 \left(\frac{2\sqrt{(1-\tau)xa}}{\tau} \right).$$

The Hille–Hardy identity, at parameter $\alpha = 0$, says

$$K_\tau(x, a) = \sum_{j=0}^{\infty} (1-\tau)^j L_j(x) L_j(a).$$

See DLMF 18.18.27. Although written using a square root, $K_\tau(\cdot, a)$ is entire, as is immediate from the power series of I_0 .

Lemma 1. *Fix $0 < \delta < L < \infty$. There are constants $C, c, \tau_0 > 0$, depending only on this interval, such that for $x, a \in [\delta, L]$, $0 < \tau < \tau_0$, and every integer $k \geq 1$,*

$$\frac{|\partial_x^k K_\tau(x, a)|}{k!} \leq C K_{4\tau}(x, a) \left[\left(\frac{C}{\sqrt{k\tau}} \right)^k + C^k \right]. \quad (1)$$

Moreover, whenever $|z| \leq c$,

$$|K_\tau(x+z, a)| \leq C K_{4\tau}(x, a) \exp(C|z|^2/\tau). \quad (2)$$

Proof. Consider first the radial Gaussian kernel

$$B_\tau(y, a) = \tau^{-1} e^{-(y+a)/\tau} I_0(2\sqrt{ya}/\tau).$$

For real positive arguments in any fixed compact interval, the elementary integral formula

$$I_0(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{t \cos \phi} d\phi$$

and $1 - \cos \phi \asymp \phi^2$ on $[-\pi, \pi]$ show that

$$I_0(t) \asymp e^t / \sqrt{t} \quad (t \geq 1), \quad B_\tau(y, a) \asymp \tau^{-1/2} e^{-(\sqrt{y}-\sqrt{a})^2/\tau}. \quad (3)$$

The constants in the second comparison are uniform on that compact interval for sufficiently small τ .

Let $y > 0$, $|z| \leq c$, and write $\sqrt{y+z} = u + iv$, choosing the branch with $u > 0$. On the relevant fixed compact intervals, choosing c small gives

$$|u - \sqrt{y}| + |v| \leq C|z|.$$

Taking absolute values inside the integral formula for B_τ gives the exact useful domination

$$|B_\tau(y+z, a)| \leq e^{v^2/\tau} B_\tau(u^2, a).$$

Since $(u - \sqrt{a})^2 \geq \frac{1}{2}(\sqrt{y} - \sqrt{a})^2 - C|z|^2$, formula (3) therefore gives

$$|B_\tau(y+z, a)| \leq C\tau^{-1/2} \exp\left[-\frac{(\sqrt{y} - \sqrt{a})^2}{2\tau} + \frac{C|z|^2}{\tau}\right]. \quad (4)$$

Now $K_\tau(x, a) = e^a B_\tau((1-\tau)x, a)$. Write $y_1 = (1-\tau)x$ and $y_4 = (1-4\tau)x$. They stay in a fixed compact positive interval, and $|\sqrt{y_1} - \sqrt{y_4}| \leq C\tau$. Thus

$$(\sqrt{y_1} - \sqrt{a})^2 \geq \frac{1}{2}(\sqrt{y_4} - \sqrt{a})^2 - C\tau^2.$$

Use this in (4), then the lower bound (3) for $B_{4\tau}(y_4, a)$. The additional factor $e^{C\tau}$ is bounded. This proves (2).

Cauchy's formula on a circle of radius $R \leq c$ consequently yields

$$\frac{|\partial_x^k K_\tau(x, a)|}{k!} \leq CK_{4\tau}(x, a) R^{-k} e^{CR^2/\tau}.$$

Choose $R = \min(c, \sqrt{k\tau/(2C)})$. If the second argument is smaller, the result is bounded by $CK_{4\tau}(C/\sqrt{k\tau})^k$. Otherwise $k\tau \geq 2Cc^2$, so $e^{Cc^2/\tau} \leq e^{k/2}$, and the result is at most $CK_{4\tau}C^k$. This proves (1). $\square$

Corollary 1 (Centered polarization error on a fixed interval). *Let $z_1, \dots, z_N \in [\delta, L]$ have mean v and range at most H . Set $a_k = e_k(z_1 - v, \dots, z_N - v) / \binom{N}{k}$. For any integer $d \leq N$, if*

$$H \leq c, \quad H/\sqrt{N\tau} \leq c,$$

then, uniformly in $a \in [\delta, L]$,

$$\left| \sum_{k=2}^d \frac{\partial_v^k K_\tau(v, a)}{k!} a_k \right| \leq CK_{4\tau}(v, a) H^2 \left(\frac{1}{N\tau} + \frac{1}{N} \right). \quad (5)$$

Constants depend only on δ, L .

Proof. The elementary centered coefficient bound gives $|a_k| \leq (CH\sqrt{k/N})^k$; a proof is in `independent_directions/marked_cancellation_identity.tex`. For the first term in (1), sum the geometric series in $CH/\sqrt{N\tau}$ beginning at $k=2$. For the second term, use $k \leq N$ to write

$$(CH)^k (k/N)^{k/2} \leq (CH)^k k/N,$$

and sum. This gives (5). $\square$

In (5) the expression is a finite differential sum, not a claim that the infinite Poisson kernel is a polynomial. Applying it to exact finite moment identities requires a justified polynomial truncation. Likewise, the constants here are not uniform as $\delta \downarrow 0$.